2.2 Measure Preliminaries-2

June 12, 2026

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Definition 0.1. If $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measurable spaces a mapping $f : X \to Y$ is called $(\mathcal{M}, \mathcal{N})$-measurable or <u>measurable</u> if $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{N}$.

Remark 0.1. Composition of measurable mappings are measurable.

Proposition 0.1. If $\mathcal{N}$ is generated by $\mathcal{E}$, then $f : X \to Y$ is $(\mathcal{M}, \mathcal{N})$-measurable iff $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$.

Proof. Only if condition is trivial. For if condition, observe $\mathcal{F} = \{E \subseteq Y \mid f^{-1}(E) \in \mathcal{M}\}$ is a σ-algebra, that contains $\mathcal{E}$ by hypothesis; it therefore contains $\mathcal{N}$. $\square$

Corollary 0.1. If X and Y are metric (or topological spaces),

Every continuous mapping is (B_X, B_Y) measurable.

* If $(X, \mathcal{M})$ is a measurable space, a real or complex value function f on X is called measurable if f is $(\mathcal{M}, B_{\mathbb{R} \text{ or } \mathbb{C}})$ measurable.